Thiago de Oliveira Sampaio

Backend Software Engineer • Java, Spring Boot & Go • Microservices • Legacy Modernization

Fortaleza, CE — Brazil | thiagooliveira1039@gmail.com | +55 85 99123-8295

linkedin.com/in/thiago-de-oliveira-sampaio-0085a8239 | github.com/Skaduhs5232 | skaduhs5232-github-io.vercel.app

PROFESSIONAL SUMMARY

Software Engineer with 5+ years of development experience (since 2021; on product teams since 2023), focused on **backend**: **Java (Spring Boot), Go, microservices, and REST/gRPC APIs**. As Senior Software Engineer at S4E, I **technically lead a team of 5 modernizing a legacy ERP** (AS-IS/TO-BE, target architecture, ADRs, API contracts). At Colégio Christus I grew from intern to Full Stack Developer II, designing microservices for **Exitus** (12 modules, about **11,000 students across 4 schools**) and shipping **LLM-based educational chatbots** with specialist agents and MCP to production. B.Sc. in Computer Science; M.Sc. candidate in Applied Informatics (UNIFOR).

PROFESSIONAL EXPERIENCE

Senior Software Engineer
Jun 2026 – Present

S4E — Fortaleza, CE (hybrid)

- **Technically lead a team of 5 modernizing a legacy ERP**, delivering incrementally alongside the running system.
- Map **AS-IS flows** with business stakeholders and design the **TO-BE scenarios** that guide the migration, making explicit the business rules that only existed in legacy code.
- Define the **target architecture** and the **API contracts** between modernized modules.
- Record technical decisions as **ADRs**, architecture diagrams, and standards guides, reducing the team's reliance on tacit knowledge.
- Run code reviews and technical mentoring, align development standards across squads, and keep business rules traceable to technical decisions.

Full Stack Software Developer II
Sep 2024 – Jun 2026

Colégio Christus — Fortaleza, CE

- Designed and built **Java (Spring Boot) and Go microservices** for **Exitus**, a multi-tenant education ecosystem with **12 integrated modules** serving about **11,000 students** at Colégio Christus and 3 other schools, with independent deployments per domain.
- Built the **educational chatbot platform** running in production on Exitus: a **BERTimbau** gatekeeper classifies each message and routes it to **specialist agents with tools and MCP servers**, with conversation auditing and cost monitoring to cut LLM spend.
- Integrated **OCR, ICR, and Text-to-Speech** models into assessment generation, reading, and grading, removing recurring manual steps.
- Co-developed **Tiptap-Exitus** with Marcelino Braga, an open-source rich-text editor library adopted as the single editor across several modules; I solely built its spell checker, a Go microservice using HunspellGo.
- Modernized legacy modules and integrated internal and external systems via REST APIs; also shipped front end in Angular, React, and TypeScript.
- Led architecture, code review, and engineering-practice initiatives with Docker, **Kubernetes**, CI/CD, Google Cloud Platform (GCP), GitFlow, and Scrum.

Full Stack Software Developer I
Oct 2023 – Sep 2024

Colégio Christus — Fortaleza, CE

- Built end-to-end features for the institutional platforms: business rules and data access in Java and SQL on the back end; screens in Angular and TypeScript on the front end.
- Maintained and evolved production features in agile cycles with the team.

Software Development Intern
Mar 2023 – Sep 2023

Colégio Christus — Fortaleza, CE

- Optimized REST APIs (Spring Boot) and PostgreSQL queries on high-volume endpoints, improving the performance of critical operations.
- Delivered new features with Angular, React, and Vue in biweekly cycles, validated directly by stakeholders.
- Hired as Full Stack Developer I at the end of the internship.

Web Developer (Freelance)
Mar 2021 – Mar 2023

Sandie — Remote

- Built and ran the company website on my own, end to end: design, front and back end, hosting, SEO, support.

TECHNICAL SKILLS

Backend: Java, Spring Boot, Go (Golang), Gin, REST APIs, gRPC, Node.js/Express, FastAPI, .NET

Architecture & Leadership: Microservices, Distributed Systems, Legacy Modernization, API Design, DDD, ADR, AS-IS/TO-BE, SOLID, Clean Code, TDD, Code Review, Technical Mentoring, Technical Leadership, Scrum

Cloud & DevOps: Docker, Kubernetes, Google Cloud Platform (GCP), CI/CD, Git, GitFlow

Data & Messaging: PostgreSQL, pgvector, MongoDB, SQL, RabbitMQ

Applied AI: LLMs, AI Agents, MCP (Model Context Protocol), RAG, LangChain, LangGraph, Hugging Face Transformers, BERTimbau, Gemini API, Anthropic API, NLP

Computer Vision & Data: YOLO (v8/v11), SAM2, OpenCV, OCR, ICR, TTS, PyTorch, TensorFlow, SciPy

Other Languages & Frontend: Python, TypeScript, JavaScript, Clojure (functional programming), C#, C; Angular, React, Svelte, HTML5, SCSS

PROJECTS, OPEN SOURCE & RESEARCH

Tiptap-Exitus — Rich-Text Editor Library (Open Source)
`TypeScript · Tiptap · Angular · Go · HunspellGo`

- Co-developed with Marcelino Braga: Tiptap components in production as the single editor across Exitus modules. I solely built the spell checker in Go with HunspellGo.

Med System
`React · Node.js · PostgreSQL · RabbitMQ · Tailwind CSS`

- Supports external physicians and Universidade Christus students in clinical care, diagnosis, and prescriptions.

B.Sc. Thesis: Asphalt Pavement Defect Detection with YOLOv11
`Python · YOLOv11X · PyTorch · SAM2 · Roboflow`

- YOLOv11X detecting potholes and patches (COCO transfer learning + data augmentation) on a custom 757-image dataset annotated with SAM2 assistance. Test set: **mAP@50 0.696** (baseline 0.628), mAP@50-95 0.503, precision 0.853.

M.Sc. Research: River-Basin Allometry + Zipf's Law in Cities
`Python · SciPy · NumPy · CCA · Geoprocessing`

- Main research on the allometry of river basins (scaling relationships in spatial data). Side article: Zipf's law in urban patches delimited with the City Clustering Algorithm (CCA), with an exponent of about 1.00 for Brazil (4,702 patches) and 0.92 for the Northeast (1,729 patches).

EDUCATION

M.Sc. in Applied Informatics
2026 – 2027 (in progress)

Universidade de Fortaleza (UNIFOR) — Research on river-basin allometry and spatial data science.

B.Sc. in Computer Science
2022 – 2025

Universidade de Fortaleza (UNIFOR) — Thesis: pavement defect detection with convolutional neural networks (YOLOv11).

PUBLICATIONS & PRESENTATIONS

SAMPAIO, T. O.; RIBEIRO, T. R. *Detecção Automatizada de Patologias em Pavimentos Asfálticos Utilizando YOLOv11* [Pavement Defect Detection with YOLOv11]. XXVI Encontro de Pós-Graduação e Pesquisa, UNIFOR, Oct 2026 (oral).

SAMPAIO, T. O.; SILVEIRA, B. L. R.; PINHEIRO, V. C. M.; COUTINHO, G. F. *Caracterização de Interações Estudante–Agente Conversacional por Processamento de Linguagem Natural* [Student–Conversational Agent Interactions via NLP]. XXVI Encontro de Pós-Graduação e Pesquisa, UNIFOR, Oct 2026 (oral).

CERTIFICATIONS & LANGUAGES

Udemy: Advanced Go (Golang) Developer (2025); Go From Scratch: Full Application; Java + Spring Boot; Machine Learning and Data Science with Python (2025); Advanced JavaScript and TypeScript; Understanding TypeScript (2025).

Languages: Portuguese (native); English (advanced).